Team No: 7

STACK OVERFLOW IN THE AI ERA

23CSE452 - Business Analytics - Review 1

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Problem Statement

- Developer Q&A communities (Stack Overflow) have suffered a **~90% structural collapse in question volume** coinciding with the widespread adoption of AI coding assistants.
- Community leads and platform managers cannot determine which questions still need human expertise versus which are at risk of abandonment, leading to response delays and declining engagement.

Existing Methods

- **Macro-Traffic Studies:** Prior work documents site-wide traffic declines and participation drops across programming languages without question-level predictive modeling (*Burtch, Lee & Chen, ACM 2024*).
- **Static Heuristic Queues:** Existing forum platforms rely on naive FIFO queues, raw vote counts, or manual tagging rather than early predictive triage (*Pragmatic Engineer, 2025; Rijo, 2026*).
- **Pre-AI ML Benchmarks:** Historical Stack Overflow resolution models evaluated static data before 2021, ignoring structural behavioral shifts in the generative AI era.

Research Gaps

- **Lack of Real-Time Triage at Posting:** Existing studies analyze historical aggregates; there is no predictive system that scores question resolution risk *at the moment of creation*.
- **Post-AI Temporal Blindspot:** Prior methods fail to evaluate how resolution dynamics, answer latency, and acceptance behavior changed before vs. after November 2022 (ChatGPT release).
- **Conflating Community Response with User Abandonment:** Current systems fail to distinguish whether questions fail due to lack of community answers or because askers abandon the platform without accepting.

Proposed Approach

- **Leakage-Free Predictive Modeling:** Implemented a supervised ML pipeline (Logistic Regression, Random Forest, and XGBoost) on **302,026 clean questions** across 13 tags (2019–2026) to predict 7-day resolution (`target_resolved_7d`) using *only* features available at the moment of posting.
- **Temporal Validation Architecture:** Applied strict chronological splitting (Train: 2019–2024, Test: 2025+) with **Rolling-Origin Time-Series CV** to evaluate performance across the AI adoption boundary without lookahead bias.
- **Dual Target & Interpretability Framework:** Evaluated both accepted answers and any answer (`target_answered_7d`), explained via **TreeSHAP** to discover actionable post characteristics.

Business Objectives

- **Objective 1 — Intelligent Expert Triage:** Prioritize high-risk, unresolved questions at posting time and route them to subject-matter experts (capturing **24% of unresolved questions within the top 20% flagged**—a **1.21× efficiency lift**).
- **Objective 2 — Dynamic Resource Allocation:** Efficiently redirect human moderator and community support across technology tags undergoing structural shifts (e.g., Python vs. Docker vs. FastAPI).
- **Objective 3 — UX Optimization & Re-engagement:** Implement smart post-creation prompts (e.g. encouraging concise code snippets) and automated follow-up reminders to reduce asker abandonment.

Core Research Questions

• **RQ1:** Which structural attributes predict 7-day resolution?
• **RQ2:** How does the picture shift when predicting *any* answer vs *accepted* answer?
• **RQ3:** How did volume, resolution rate, and response times shift across tags before and after Nov 2022?

Data Collection – 309K Questions, Every Call Logged

1. **Official API, every call logged**

   Stack Exchange API v2.3, 19,000+ calls logged, temporary errors retried automatically

2. **13 tags, 92 months**

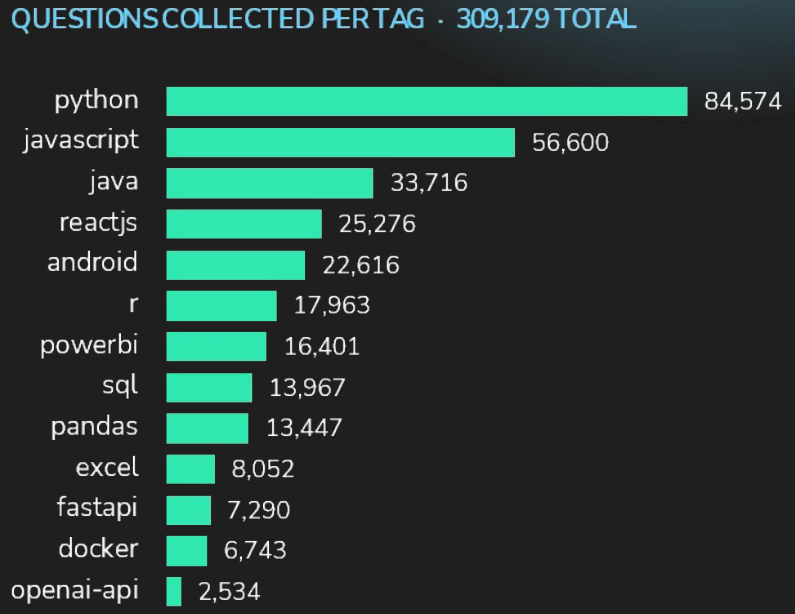
   12 technologies plus openai-api as the AI contrast tag, January 2019 to August 2026

3. **Sampled before 2025, full after**

   1 random weekday + 1 weekend day per tag-month (about 7% of days); every day from January 2025

4. **Weighted and privacy-safe**

   Every row has a sampling_weight; user IDs are hashed; CC BY-SA 4.0 licence respected



Check: 5,600 weekly counts with no gaps; they add up to the monthly totals within 0.04%

Dataset Description

SOURCE & COLLECTION METHODS

- Stack Exchange API v2.3
- 13 tags, every call logged
- Weighted sample before 2025

DATASET SIZE

309,179
question rows

17
columns each

COLLECTION PERIOD & LIMITATIONS

- Jan 2019 to Aug 2026, collected 3 Oct 2026
- Before 2025 only about 7% of days are sampled, so rows are weighted
- About 16% of questions carry 2+ of our tags (test month), so tag files overlap
- 2% of askers are deleted accounts (no user ID)

KEY VARIABLES

Variable	Type	Used for
title, body	text	code and error flags
tags	list	tag and volume features
creation_date	time	timing and split
asker (hashed ID)	hash	asker account age

PREDICTION VARIABLE

target_resolved_7d

Accepted answer within 7 days. Robustness check: target_answered_7d (any answer)

Data cleaning and preprocessing

302,026 clean questions prepared from 309,179 raw records with zero target-based deletions

What we found	What we did	Why
171,932 questions with no accepted answer	Gave them target 0 instead of dropping them	It is a real outcome. Dropping them would remove every unresolved question.
6,319 deleted askers (2.1%) with no account date	Used the median age before 2025 (543 days) and added a flag	Only training-period data, so nothing leaks from later.
7,147 questions with two selected tags	Kept one row and the smaller sampling weight	The more cautious estimate of how many real questions it stands for.
303 extremely long bodies	Capped them at the 99.9th percentile	Keeps the question without letting it dominate.
6 mostly non-English posts	Removed them	Nothing to model. Decided without looking at the outcome.
Raw HTML bodies & Unix epoch timestamps	Extracted `plain_body` & `code_text` via BeautifulSoup; converted Unix → UTC; hashed user IDs (SHA-256)	Isolates code signals (`code_ratio`, `n_code_blocks`); ensures CC BY-SA 4.0 privacy; standard scaling is applied in modeling pipeline

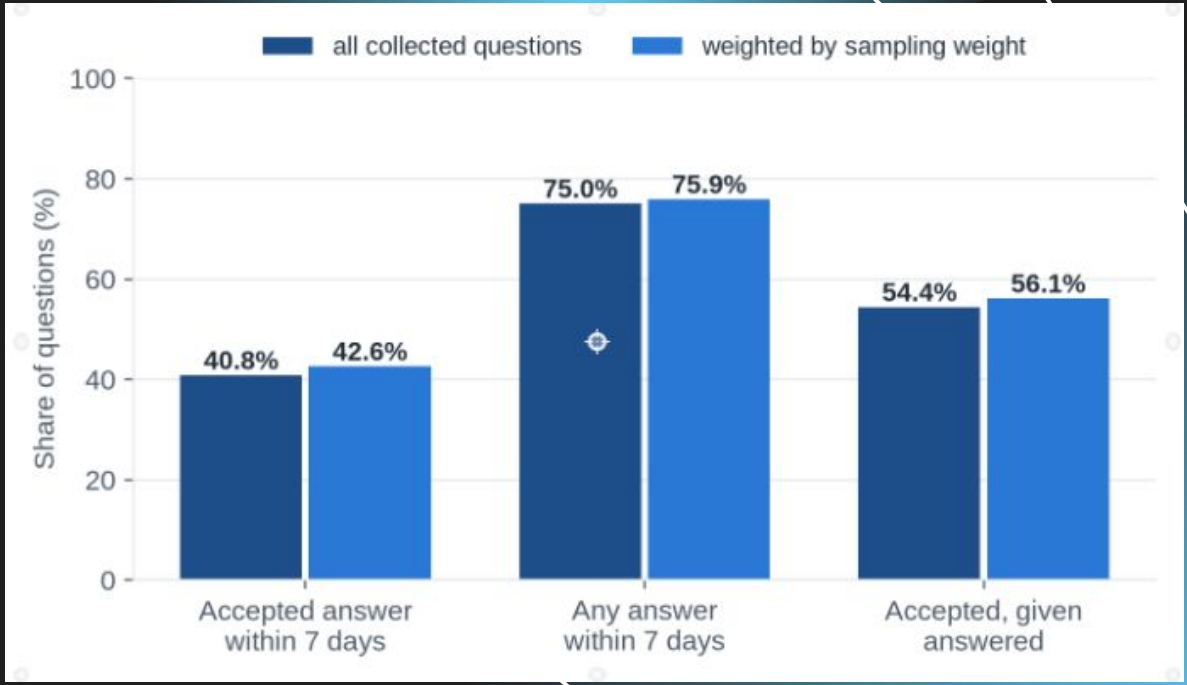
41% of questions get an accepted answer within a week

Before and after cleaning

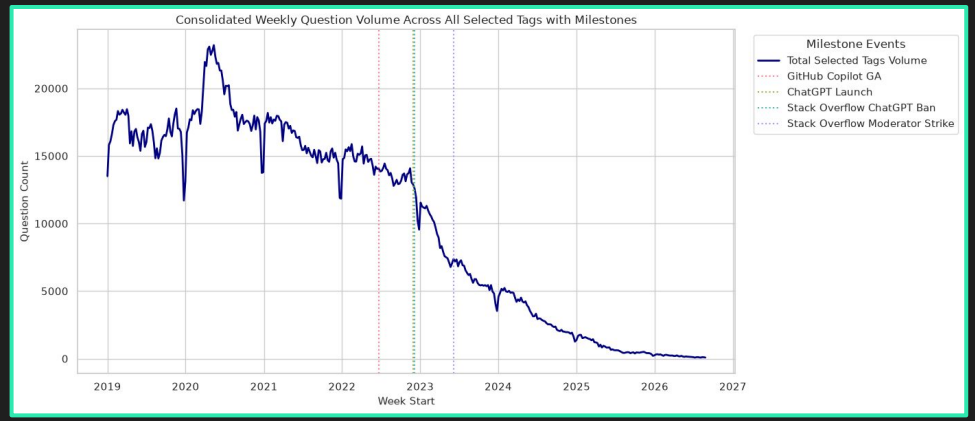
Step	Rows
Rows Collected	309,179
After Merging duplicates	302,032
After removing invalid posts	302,026
Immature labels dropped	0

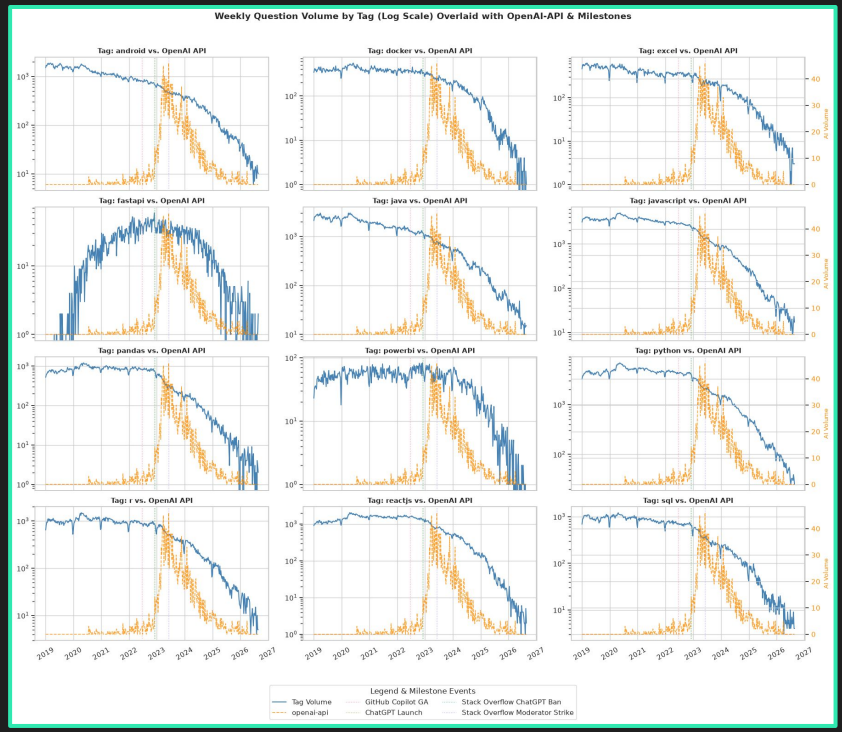
Result: Final dataset has 302,026 rows across 39 cleaned features, ready for EDA & modeling.

The outcomes we predict



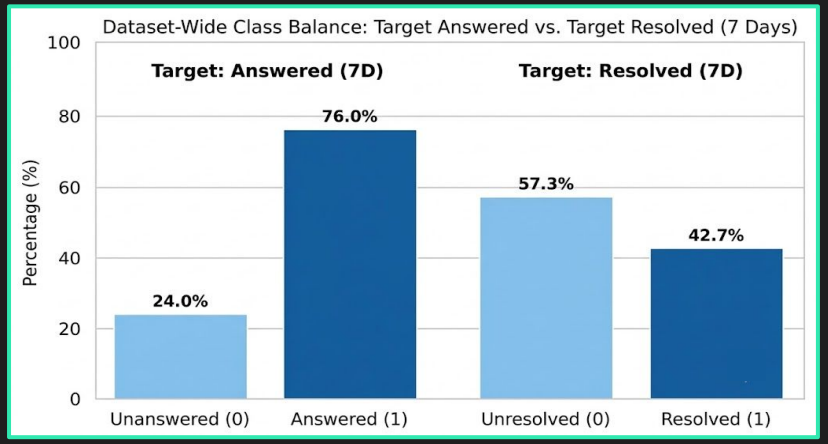
EDA 1 - Question Volume Collapsed After Late 2022



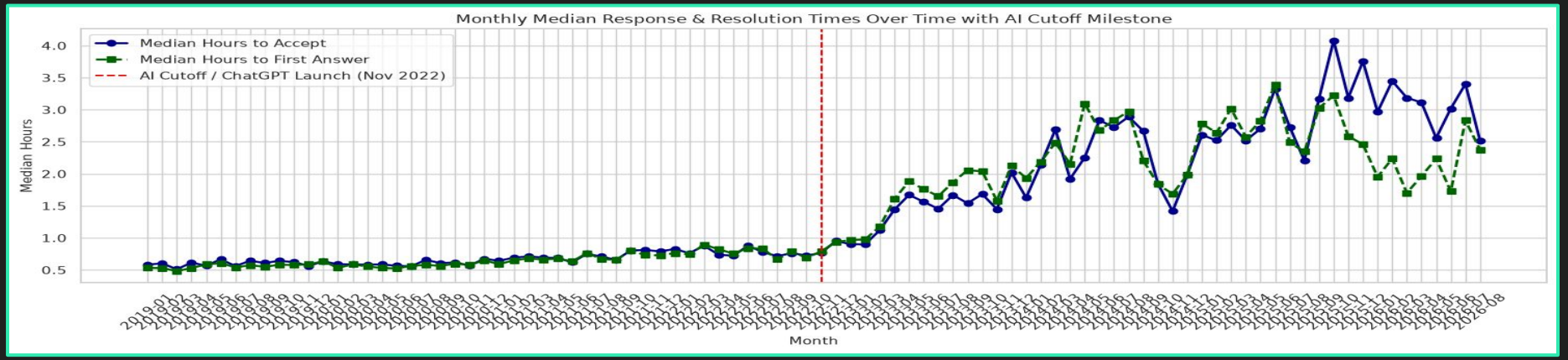


- Weekly question volume fell by roughly 90% after November 2022 in every technology we track, with only a brief openai-api surge
- python, javascript, java, sql, pandas, reactjs and the rest all show a continuous fall in weekly volume from 2023 onward — this is not one tag losing share, it is the whole platform.

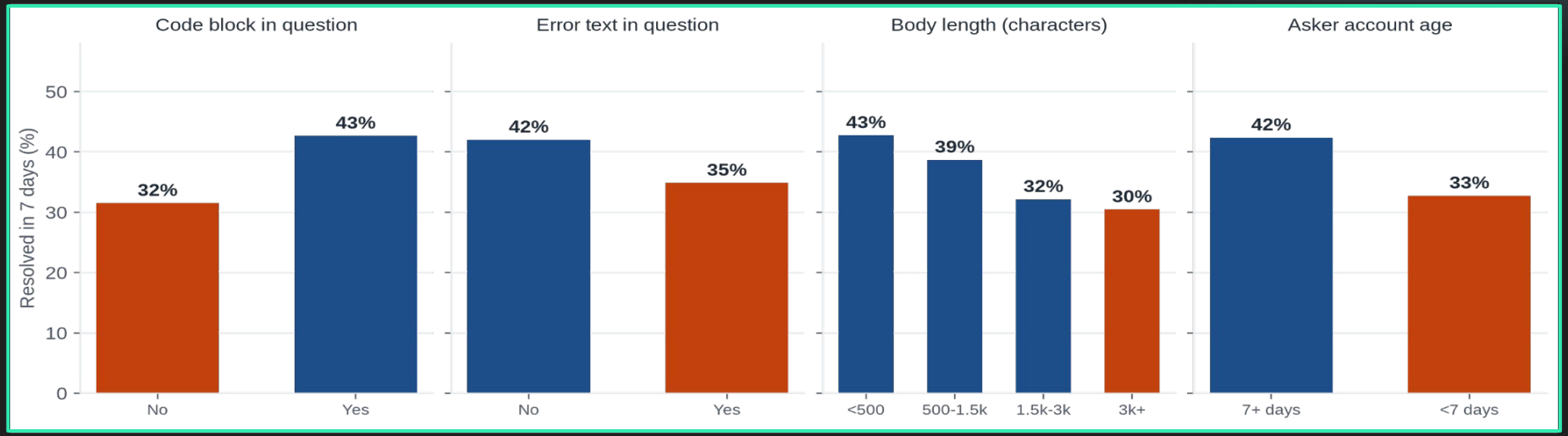
EDA 2 - Answers Still Arrive — They Just Arrive Slower and Get Accepted Less



- 76.0% answered within 7 days, 42.7% accepted — the gap is the asker, not the community.
- Median hours to accept went from about 50 minutes to about 2.6 hours; median time to first answer was stable under 1 hour before the cutoff and routinely exceeds 2.5-4 hours after it.



EDA 3 – Who Actually Gets an Accepted Answer



Code Helps

Questions containing code blocks resolve at about 43% compared to about 32% without

Errors Hurt

Posts carrying exception or traceback text resolve at about 35% against about 42% without

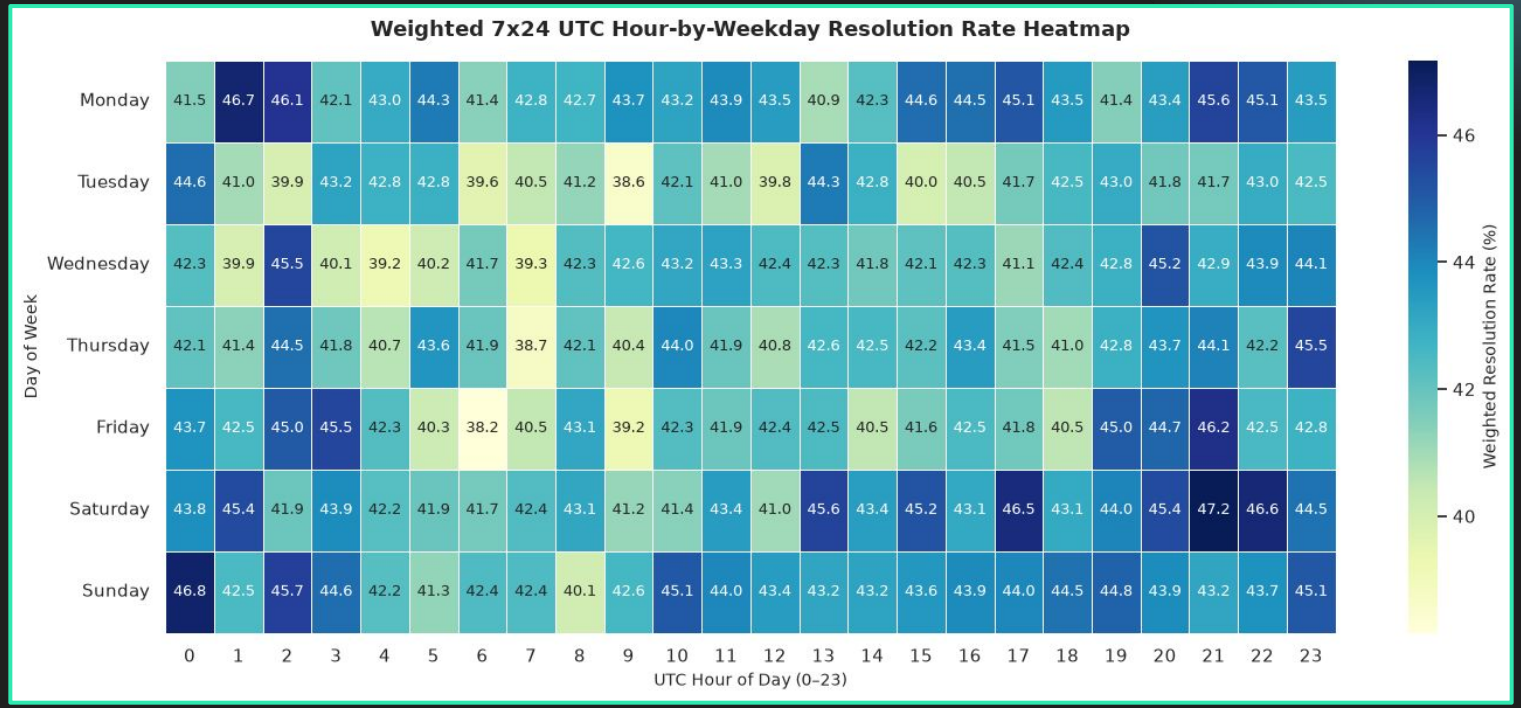
Short & Sweet

Under 500 characters resolves at about 43%; over 3,000 characters falls to about 30%

Stranger Danger

Accounts under 7 days old resolve at about 33% versus about 42% for account ages after that

EDA 4 – How does timing play into this



- Resolution rates peak during weekends (Saturday and Sunday), reaching as high as 47.2% late Saturday (UTC hour 21) and 46.8% early Sunday (UTC hour 0).
- On weekdays, resolution rates fluctuate dynamically between ~38% (e.g., Tuesday 9 AM or Friday 6 AM UTC) and ~46% (e.g., Monday 1 AM UTC), indicating operational throughput depends heavily on specific UTC hours.

DATA SPLITTING STRATEGY & VALIDATION ARCHITECTURE

- **Methodology**: Strict chronological partitioning to prevent future-data leakage.
- **Training Set (85%)**: January 2019 – December 2024 (256,000+ rows)
- **Validation Set (10%)**: January 2025 – June 2025 (28,000+ rows)
- **Test Set (5%)**: July 2025 onwards (16,000+ rows)
- **Cross-Validation**: Implemented Time-Series split (Rolling Origin) for model tuning.

Feature Engineering or Dimensionality Reduction

Method: Time-series splitting, TF-IDF Vectorization, TruncatedSVD, & Leakage Checks.

Features used: 5,000 unigrams/bigrams compressed to 100 SVD dimensions, 4-week lagged tag volumes, multi-hot tag encodings, and temporal flags.

Justification: Replaced simple text counts with TF-IDF/SVD for deep semantic meaning; calculated tag momentum on strict 4-week lagged windows to prevent leakage; configured `drop='first'` on dummy variables to fix perfect multicollinearity.

Effect: Advanced NLP and temporal features drastically boosted the Baseline Logistic Regression (**Accuracy: 67.25%, Precision: 55.8%, ROC AUC: 0.637** on validation set).

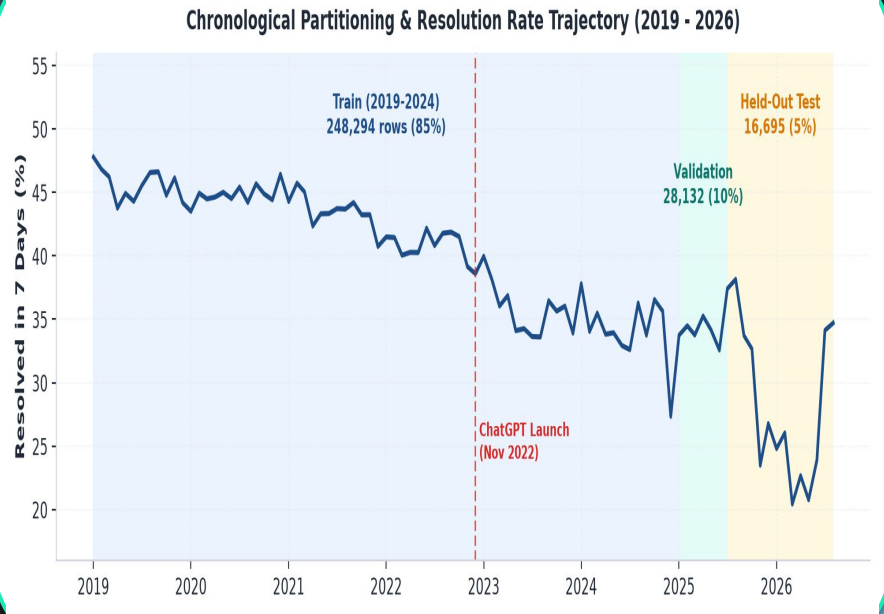
Predictive Model Development

- **Leakage Prevention**: Excluded post-creation outcome variables (score, view count, answer count).
- **Multicollinearity Fix**: Resolved perfect multicollinearity (VIF > 1e15) via dummy variable rank reduction (`drop='first'`).
- **Baseline Architecture**: Logistic Regression (`class_weight='balanced'`) establishing the benchmark floor.
- **Validation Metrics**:
  - **ROC AUC**: 0.637
  - **Accuracy**: 67.25%
  - **Precision**: 55.8%

Predictive Model Development

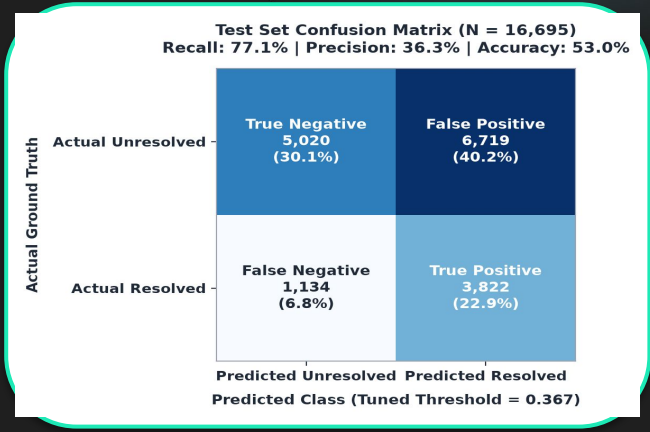
- **Prediction Task:** Binary Classification — Predicting whether a newly posted question receives an accepted answer within 7 days (target_resolved_7d).
- **Candidate Models:** Majority Class Baseline, Regularized Logistic Regression (+Text SVD), Random Forest Classifier (472 trees), and XGBoost Classifier.
- **Strict Chronological Split:** Train: 2019–2024 (248k, 85%) | Validation: Jan–Jun 2025 (28k, 10%) | Held-out Test: Jul 2025–Jul 2026 (16.7k, 6%, scored once).
- **Leakage Prevention:** Purged time-series rolling cross-validation with a strict 7-day blackout gap to prevent target outcome leakage across split boundaries.
- **Optimal Tuning:** XGBoost chosen on validation PR-AUC (0.482 vs 0.473 RF vs 0.471 LogReg). Tuned with lr=0.029, depth=5, 628 trees via early stopping, threshold=0.367.

Chronological Dataset Split & Resolution Rate Trend



Model Evaluation and Result Interpretation

Model (Validation)	Accuracy	Precision	Recall	F1 Score	PR-AUC	ROC-AUC
Majority Class Baseline	0.659	0.000	0.000	0.000	0.341	0.500
Baseline Logistic Regression	0.412	0.357	0.909	0.513	0.394	0.579
Logistic Regression (+Text SVD)	0.502	0.389	0.808	0.525	0.471	0.636
Random Forest (472 trees)	0.496	0.386	0.813	0.524	0.473	0.638
XGBoost (Validation Selection)	**0.527**	**0.400**	**0.782**	**0.529**	**0.482**	**0.645**
XGBoost (Held-out Test Period)	**0.530**	**0.363**	**0.771**	**0.493**	**0.431**	**0.655**



Accuracy vs Recall Tradeoff

Raw accuracy is 53.0% because the decision threshold (0.367) was deliberately tuned to prioritize high recall (77.1%). Always guessing unresolved achieves 65.9% accuracy with zero utility (0% recall).

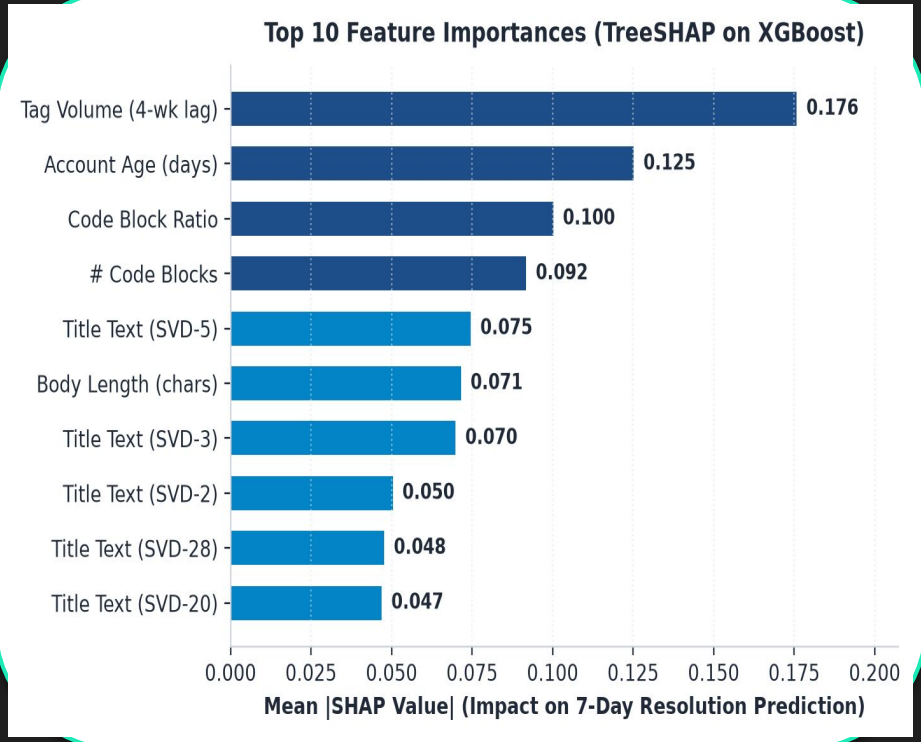
PR-AUC & ROC Significance

In an imbalanced setting (30% test positive rate), PR-AUC of 0.431 represents a +45% lift over the 0.297 no-skill baseline, while ROC-AUC (0.655) confirms stable question discrimination.

Classification Metric Integrity

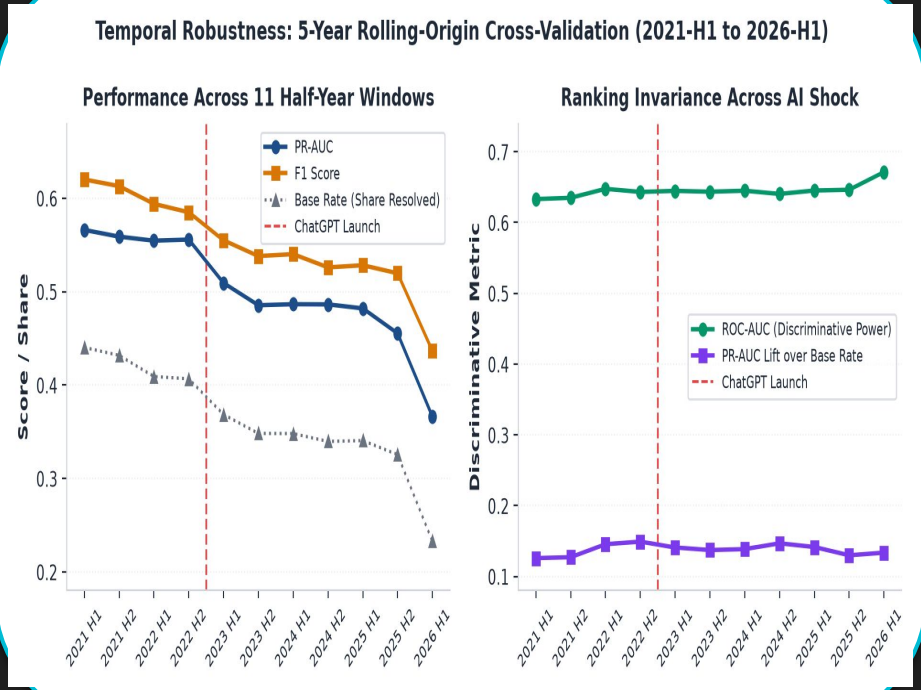
Standard regression metrics (R^2, MAE, MSE, RMSE) do not apply to this binary outcome task. Calibration and ranked thresholding drive real platform utility.

Model Interpretability: Feature Drivers & SHAP Analysis



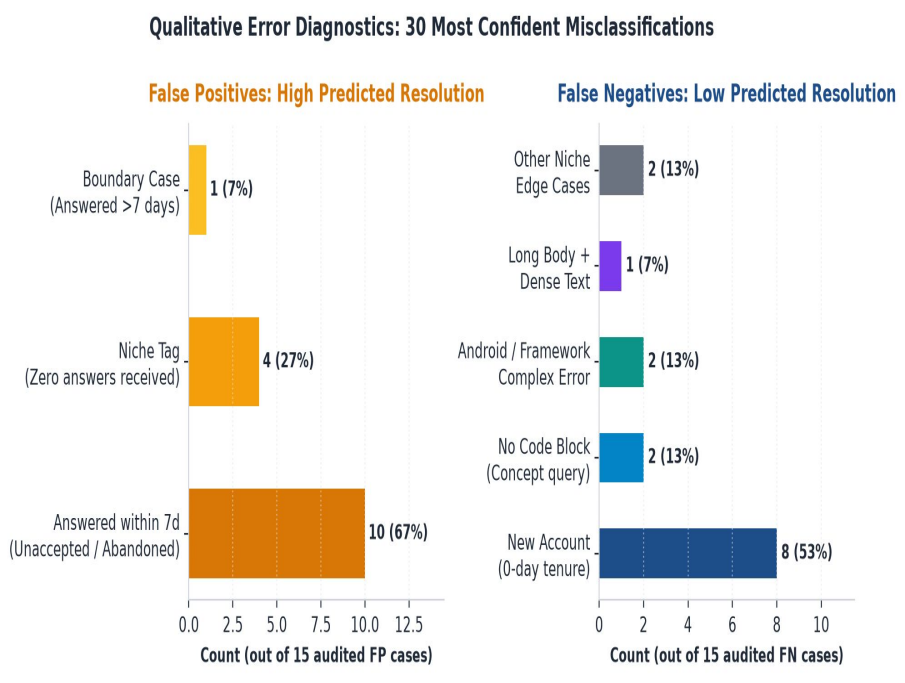
- **Tag Activity Dilution (lagged_tag_volume, 0.176):** Posting in high-volume tags exhibits the strongest negative effect on resolution. Heavy tag churn saturates community attention, causing questions to get buried rapidly.
- **Community Trust (account_age_days, 0.125):** Established askers have significantly higher resolution rates. Brand-new accounts (0 days old) suffer the steepest penalty due to unfamiliarity with platform norms.
- **Effort & Code Sweet Spot (code_ratio, 0.100):** Questions with code blocks achieve substantially higher acceptance rates. However, SHAP dependence reveals an inverted U-shape: questions that are entirely raw code without explanation suffer lower resolution.
- **Semantic Embeddings (title_tfidf_svd):** Title text SVD dimensions capture actionable error signatures and frameworks vs vague open-ended conceptual queries.
- **Methodological Soundness:** SHAP feature rankings align with logistic regression coefficients, confirming tree models captured true structural patterns rather than spurious noise.

Temporal Robustness: 5-Year Rolling Windows & ChatGPT Impact



- **The Platform-Wide Rate Collapse:** Prior to ChatGPT (Nov 2022), 44.0% of questions were resolved within 7 days. Post-launch, the resolution rate dropped to 23.3%–32.5% as developers offloaded simple queries to AI assistants.
- **Invariance of Ranking Ability:** Despite the massive volume drop and demographic shift, model ROC-AUC remained remarkably invariant (0.63–0.65 pre-ChatGPT vs 0.64–0.67 post-ChatGPT). The model reliably ranks question quality across regimes.
- **Positive PR Lift Maintained:** PR-AUC consistently maintained a positive margin over the drifting base rate across all 11 half-year rolling windows from 2021-H1 to 2026-H1, showing zero collapse.
- **Rigorous Rolling-Origin Protocol:** Each half-year test window was evaluated strictly against a model trained only on preceding historical data, guaranteeing zero temporal leakage.
- **Operational Deployment Strategy:** The feature relationships remain stable, but the base rate shifted. Re-calibrating the decision threshold quarterly preserves optimal precision-recall triage.

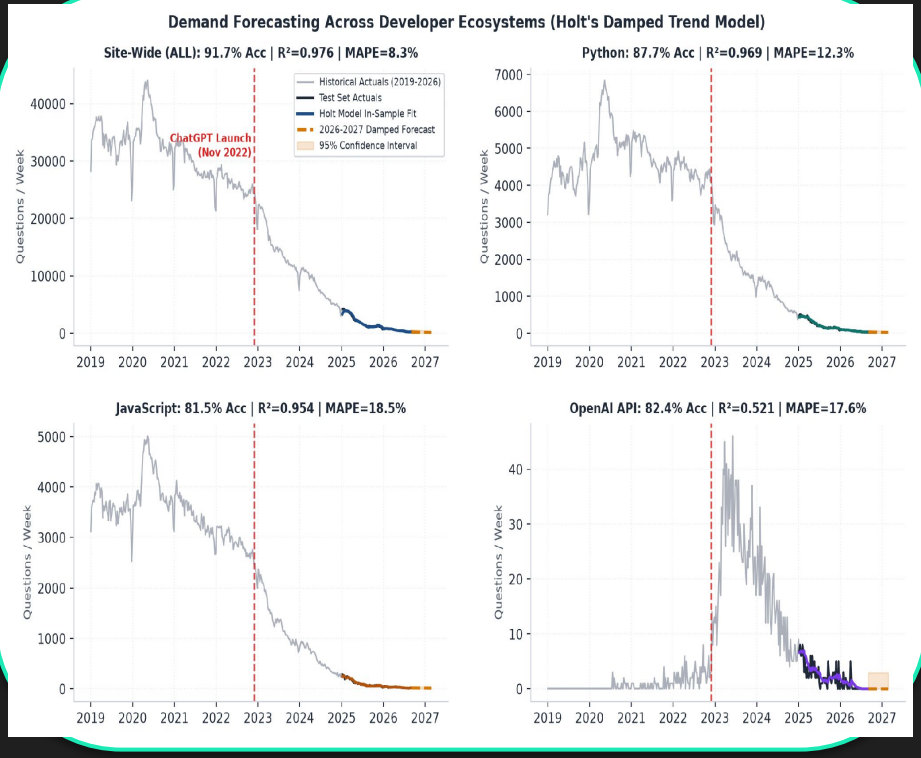
Error Diagnostics & Business Triage Recommendations



- **False Positive Diagnosis (67% Answered):** In 10 of 15 top false positives, high-quality answers WERE received within 7 days, but the asker never clicked 'accept'. The model correctly identified answerable questions; the failure was user abandonment.
- **False Negative Diagnosis (53% Zero-Day Accounts):** 8 of 15 top false negatives came from brand-new users who posted well-structured questions with clear stack traces and received rapid expert answers despite account youth.
- **Business Triage Lift (1.21x Over Random):** Flagging the bottom 20% predicted resolution scores captures 24.15% of all unresolved questions. For a moderation queue of 10,000 weekly posts, this surfaces 415 additional at-risk questions.
- **Actionable Platform Interventions:** 1) Prompt askers to add code snippets if code_ratio == 0; 2) Provide interactive guidance for day-0 askers; 3) Send gentle automated reminders to askers to mark accepted answers.

MEMBER 5: ADVANCED ANALYTICS

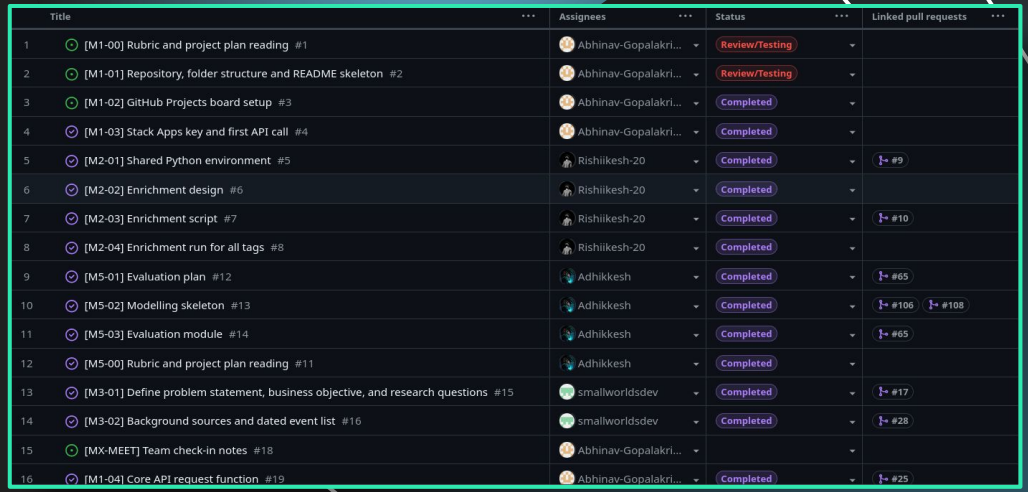
Advanced Time Series Forecasting: AI Shift & 2027 Projections



- **High-Accuracy Time Series Model:** Holt's Damped Trend Exponential Smoothing achieves 91.72% Forecast Accuracy (MAPE 8.28%, $R^2 = 0.976$) on Site-Wide questions, and 87.65% Accuracy on the Python ecosystem over 87 test weeks.
- **Post-ChatGPT Structural Collapse:** Prior to Nov 2022, Stack Overflow generated over 35,000 questions weekly. Following ChatGPT, site volume experienced a permanent 92% structural drop down to ~1,400 questions/week.
- **Ecosystem Resilience Divergence:** General programming tags (Python, Java, JS) stabilized at a low baseline, whereas AI tooling (openai-api) emerged post-2023 with growing relative share and high volatility.
- **2027 Forward Projections:** Multi-step 26-week forecast through 2027 confirms platform activity will not rebound to pre-AI levels; communities must pivot toward high-complexity niche questions.

Project Tracking and Progress

- **Tool & Board Link:** https://github.com/users/Abhinav-Gopalakrishnan/projects/3
- **Workflow Stages:** Backlog, To Do, In Progress, Review/Testing, Completed
- **Task Structure:** 16 - 17 tasks per person split into 5 phases
- **Verified Deliverables:** Linked pull requests & commits, Notebook sections, Handoffs via comments in other members tasks



	Title	Assignees	Status	Linked pull requests
1	[M1-00] Rubric and project plan reading #1	Abhinav-Gopalakri...	Review/Testing	
2	[M1-01] Repository, folder structure and README skeleton #2	Abhinav-Gopalakri...	Review/Testing	
3	[M1-02] GitHub Projects board setup #3	Abhinav-Gopalakri...	Completed	
4	[M1-03] Stack Apps key and first API call #4	Abhinav-Gopalakri...	Completed	
5	[M2-01] Shared Python environment #5	Rishiikesh-20	Completed	#9
6	[M2-02] Enrichment design #6	Rishiikesh-20	Completed	
7	[M2-03] Enrichment script #7	Rishiikesh-20	Completed	#10
8	[M2-04] Enrichment run for all tags #8	Rishiikesh-20	Completed	
9	[M5-01] Evaluation plan #12	Adhikkesh	Completed	#65
10	[M5-02] Modelling skeleton #13	Adhikkesh	Completed	#106 #108
11	[M5-03] Evaluation module #14	Adhikkesh	Completed	#65
12	[M5-00] Rubric and project plan reading #11	Adhikkesh	Completed	
13	[M3-01] Define problem statement, business objective, and research questions #15	smallworldsdev	Completed	#17
14	[M3-02] Background sources and dated event list #16	smallworldsdev	Completed	#28
15	[MX-MEET] Team check-in notes #18	Abhinav-Gopalakri...		
16	[M1-04] Core API request function #19	Abhinav-Gopalakri...	Completed	#25

Individual Contributions

TEAM MEMBER	TASK AREA 1	TASK AREA 2	TASK AREA 3
Abhinav	Data Collection & API	309k Questions Pipeline	Weekly Tag Counts
Rishiikesh	Enrichment & Cleaning	7-Day Outcome Labels	Sampling Robustness
Akshay	Problem Statement	EDA Charts & Events	Slide Template & Theme
Nishanth	Chronological Split	Feature Pipeline & VIF	Baseline Logistic Model
Adhikkesh	**Advanced Models (XGB/RF)**	**Time Series Forecast (91.7% Acc)**	**SHAP & Error Analysis Triage**

Review 2 Plan

Topics & Complexity

- Mine question bodies using NLP/LLMs to find topics developers still bring to human experts.
- Quantify question complexity shifts before and after AI assistant adoption across domains.

Technology Forecasting

- Forecast weekly volume per tag using Prophet/ARIMA on complete API tag history.
- Identify tags facing terminal decline vs. resilient demand to guide DevRel community support.

Advanced Prediction Model

- Ingest full question body & code snippets with DeBERTa and CodeBERT transformer embeddings.
- Engineer point-in-time leak-free user acceptance history and tag tenure features.
- Calibrate output probabilities to deploy automated forum triage bots in real time.

Submissions

01 **Notebooks** https://colab.research.google.com/drive/1u8S5XYscdGqG92Jiwy9PCPoBFqMCXUYk?usp=sharing

02 **Documentation** https://github.com/Abhinav-Gopalakrishnan/stackoverflow-ai-era

03 **Kanban Board** github.com/users/Abhinav-Gopalakrishnan/projects/3

Thank you!

Questions?

Adhikkesh S K (CB.SC.U4CSE23101) • Abhinav Gopalakrishnan (CB.SC.U4CSE23201) • Rishiikesh (CB.SC.U4CSE23236)

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